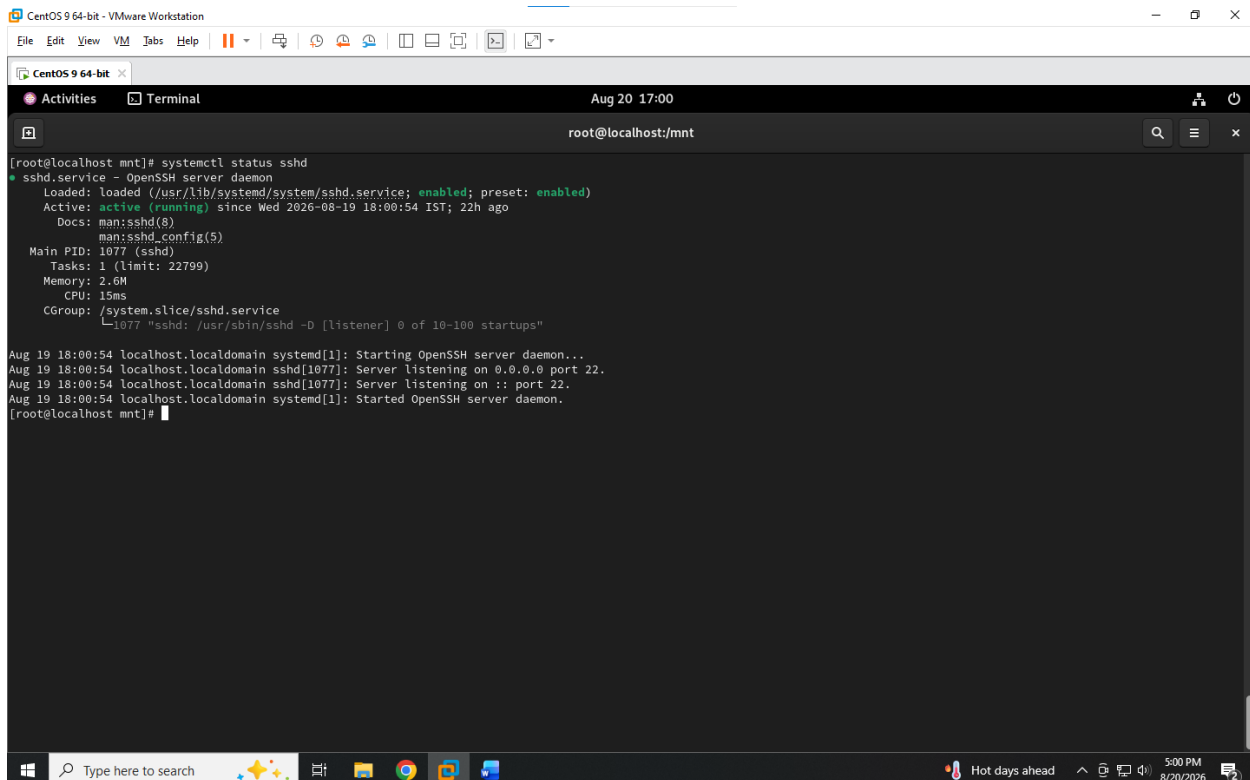


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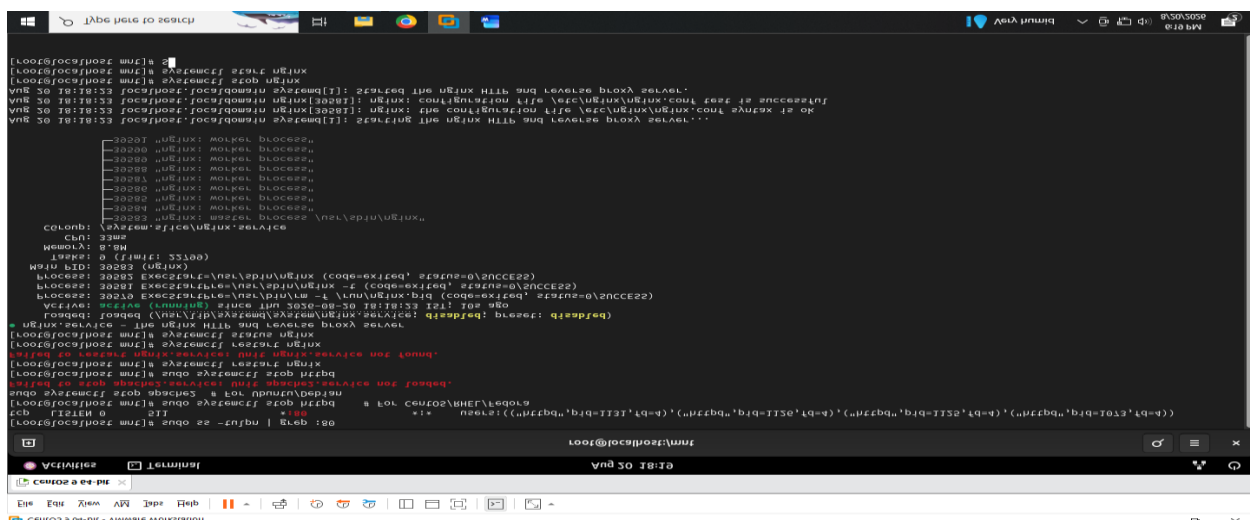
Q.1 Use the systemctl command to check the status of the sshd service on your Linux system and write down whether it is active, inactive, or failed.

ANS : As we can see in this img the ssh is open and working .



Q.2 Start the nginx service using systemctl, then immediately stop it. After each action, verify the status and note the changes. Hint : Use 'sudo systemctl start nginx' and 'sudo systemctl stop nginx'.

ANS :

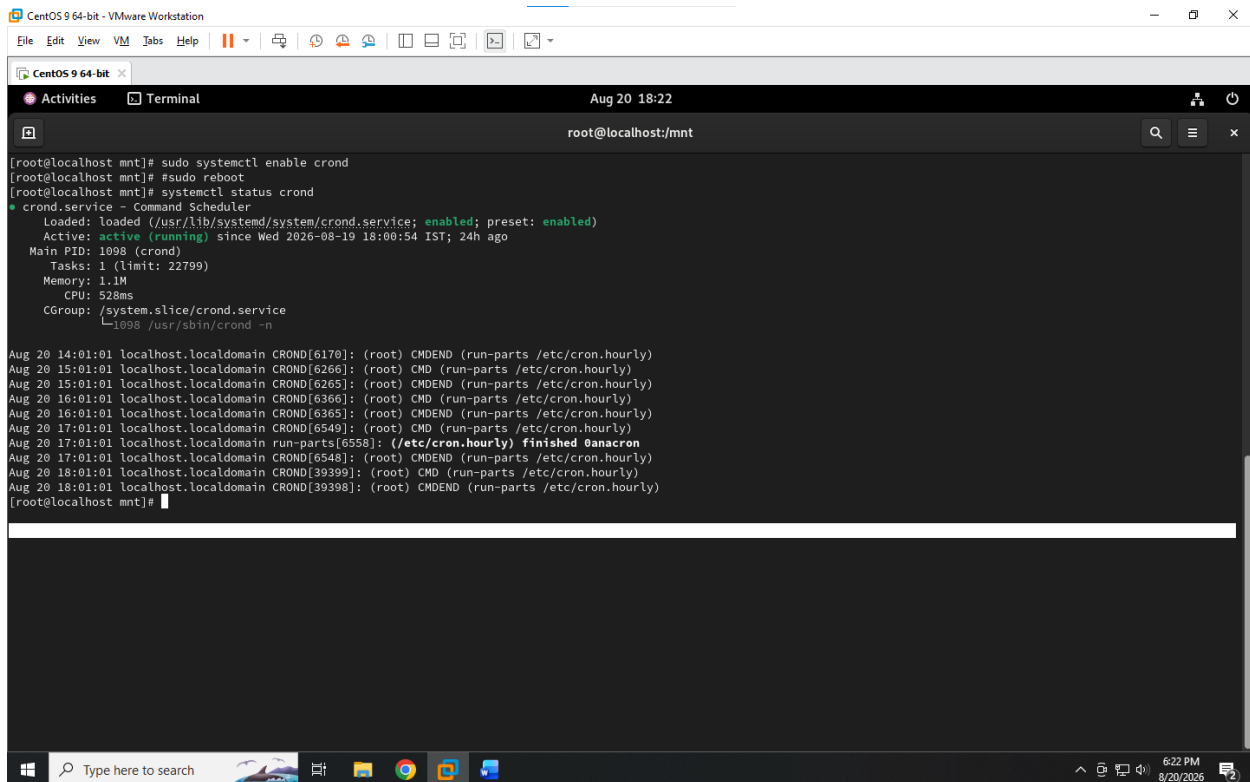


## MANAGE INSTALLED SERVICES

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Q.3 Enable the cron service to start automatically on boot, then reboot your system and confirm that the service is running after reboot.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 20 18:22
root@localhost:/mnt

[root@localhost mnt]# sudo systemctl enable crond
[root@localhost mnt]# #sudo reboot
[root@localhost mnt]# systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-19 18:00:54 IST; 24h ago
     Main PID: 1098 (crond)
       Tasks: 1 (limit: 22799)
      Memory: 1.1M
         CPU: 528ms
    CGroup: /system.slice/crond.service
            └─1098 /usr/sbin/crond -n

Aug 20 14:01:01 localhost.localdomain CROND[6170]: (root) CMDEND (run-parts /etc/cron.hourly)
Aug 20 15:01:01 localhost.localdomain CROND[6266]: (root) CMD (run-parts /etc/cron.hourly)
Aug 20 15:01:01 localhost.localdomain CROND[6265]: (root) CMDEND (run-parts /etc/cron.hourly)
Aug 20 16:01:01 localhost.localdomain CROND[6366]: (root) CMD (run-parts /etc/cron.hourly)
Aug 20 16:01:01 localhost.localdomain CROND[6365]: (root) CMDEND (run-parts /etc/cron.hourly)
Aug 20 17:01:01 localhost.localdomain CROND[6549]: (root) CMD (run-parts /etc/cron.hourly)
Aug 20 17:01:01 localhost.localdomain run-parts[6558]: (/etc/cron.hourly) finished 0anacron
Aug 20 17:01:01 localhost.localdomain CROND[6548]: (root) CMDEND (run-parts /etc/cron.hourly)
Aug 20 18:01:01 localhost.localdomain CROND[39399]: (root) CMD (run-parts /etc/cron.hourly)
Aug 20 18:01:01 localhost.localdomain CROND[39398]: (root) CMDEND (run-parts /etc/cron.hourly)
[root@localhost mnt]#
```

Q.4 List all currently running services using systemctl and identify any service related to networking (like NetworkManager or networking). Write the current status of that service.

ANS :

# MANAGE INSTALLED SERVICES

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```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal
root@localhost:/mnt — systemctl list-units --type=service --state=running
Aug 20 18:24

UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service             loaded active running Accounts Service
alsa-state.service                  loaded active running Manage Sound Card State (restore and store)
atd.service                          loaded active running Deferred execution scheduler
auditd.service                     loaded active running Security Auditing Service
avahi-daemon.service                loaded active running Avahi mDNS/DNS-SD Stack
chronyd.service                     loaded active running NTP client/server
colord.service                       loaded active running Manage, Install and Generate Color Profiles
crond.service                       loaded active running Command Scheduler
cups.service                         loaded active running CUPS Scheduler
dbus-broker.service                 loaded active running D-Bus System Message Bus
firewalld.service                   loaded active running FirewallD - dynamic firewall daemon
fwupd.service                       loaded active running Firmware update daemon
gdm.service                         loaded active running GNOME Display Manager
irqbalance.service                 loaded active running irqbalance daemon
libstoragemgmt.service              loaded active running libstoragemgmt plug-in server daemon
mclog.service                       loaded active running Machine Check Exception Logging Daemon
ModemManager.service                loaded active running Modem Manager
NetworkManager.service              loaded active running Network Manager
nginx.service                       loaded active running The nginx HTTP and reverse proxy server
pcscd.service                       loaded active running PC/SC Smart Card Daemon
polkit.service                      loaded active running Authorization Manager
power-profiles-daemon.service         loaded active running Power Profiles daemon
rsyslog.service                     loaded active running system Logging Service
rtkit-daemon.service                loaded active running RealtimeKit Scheduling Policy Service
sshd.service                        loaded active running OpenSSH server daemon
sssd-kcm.service                    loaded active running SSSD Kerberos Cache Manager
switcheroo-control.service           loaded active running switcheroo Control Proxy service
systemd-journald.service             loaded active running Journal Service
systemd-logind.service               loaded active running User Login Management
systemd-udevd.service                loaded active running Rule-based Manager for Device Events and Files
udisks2.service                     loaded active running Disk Manager
upower.service                      loaded active running Daemon for power management
userd.service                       loaded active running User Manager for UID 0
vgauthd.service                     loaded active running VGAuthService for open-vm-tools
vmtoolsd.service                     loaded active running Service for virtual machines hosted on VMware
wpa_supplicant.service               loaded active running WPA supplicant

LOAD = Reflects whether the unit definition was properly loaded.
[root@localhost /mnt]#
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal
root@localhost:/mnt — systemctl status NetworkManager
Aug 20 18:25

switcheroo-control.service           loaded active running Switcheroo Control Proxy service
systemd-journald.service             loaded active running Journal Service
systemd-logind.service               loaded active running User Login Management
systemd-udevd.service                loaded active running Rule-based Manager for Device Events and Files
udisks2.service                     loaded active running Disk Manager
upower.service                      loaded active running Daemon for power management
userd.service                       loaded active running User Manager for UID 0
vgauthd.service                     loaded active running VGAuthService for open-vm-tools
vmtoolsd.service                     loaded active running Service for virtual machines hosted on VMware
wpa_supplicant.service               loaded active running WPA supplicant

LOAD = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
SUB = The low-level unit activation state, values depend on unit type.
36 loaded units listed.
[root@localhost /mnt]# systemctl list-units --type=service --state=running | grep -E "NetworkManager|network"
NetworkManager.service              loaded active running Network Manager
[root@localhost /mnt]# systemctl status NetworkManager
NetworkManager.service - Network Manager
Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: enabled)
Active: active (running) since Thu 2026-08-20 17:13:32 IST; 1h 12min ago
Docs: man:NetworkManager(8)
Main PID: 7525 (NetworkManager)
Tasks: 3 (limit: 22799)
Memory: 7.5M
CPU: 229ms
CGroup: /system.slice/NetworkManager.service
└─7525 /usr/sbin/NetworkManager --no-daemon

Aug 20 17:17:56 localhost.localdomain NetworkManager[7525]: <info> [1787226476.5318] device (ens160): state change: secondaries -> activated (reason 'none', managed-type: 'full')
Aug 20 17:17:56 localhost.localdomain NetworkManager[7525]: <info> [1787226476.5321] manager: NetworkManager state is now CONNECTED_SITE
Aug 20 17:17:56 localhost.localdomain NetworkManager[7525]: <info> [1787226476.5323] device (ens160): Activation: successful, device activated.
Aug 20 17:17:56 localhost.localdomain NetworkManager[7525]: <info> [1787226476.5325] manager: NetworkManager state is now CONNECTED_GLOBAL
Aug 20 17:32:56 localhost.localdomain NetworkManager[7525]: <info> [1787227376.5234] dhcp4 (ens160): state changed new lease, address=192.168.31.130
Aug 20 17:36:39 localhost.localdomain NetworkManager[7525]: <info> [1787227599.9321] agent-manager: agent[8b25f4566ba89b56;1.69/org.gnome.Shell.NetworkAgent/0]: agent registered
Aug 20 17:47:56 localhost.localdomain NetworkManager[7525]: <info> [1787228276.5283] dhcp4 (ens160): state changed new lease, address=192.168.31.130
Aug 20 18:02:56 localhost.localdomain NetworkManager[7525]: <info> [1787229176.5248] dhcp4 (ens160): state changed new lease, address=192.168.31.130
Aug 20 18:15:48 localhost.localdomain NetworkManager[7525]: <info> [1787229948.4121] agent-manager: agent[a97530ad4dc15a65;1.69/org.gnome.Shell.NetworkAgent/0]: agent registered
Aug 20 18:17:56 localhost.localdomain NetworkManager[7525]: <info> [1787230076.5254] dhcp4 (ens160): state changed new lease, address=192.168.31.130
lines 1-21/21 (END)
```

Q.5 Simulate a real-world scenario: Imagine you are troubleshooting why Spotify Web cannot stream music on your Linux machine. Use systemctl to check if the firewall (ufw or firewalld) service is running, and if it is, disable it temporarily. Document each command used and the result. Constraint: Only disable the firewall if you are on a safe, non-production environment.

ANS : Here is the expanded output detailing the full simulation process with plain text commands:

## 1. Identify the Active Firewall Service

Before taking action, check which firewall daemon is active on the host machine.

Command:

```
sudo systemctl status firewalld
```

Simulated Output:

firewalld.service - firewalld - dynamic firewall daemon

Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; vendor preset: enabled)

Active: active (running) since Thu 2026-08-20 18:00:00 IST; 25min ago

Main PID: 852 (firewalld)

## 2. Inspect Open Network Connections

Check active listening ports and established web traffic connections to see if browser sockets are being restricted.

Command:

```
sudo ss -tulpn | grep -E "80|443"
```

Simulated Output:

```
tcp LISTEN 0 128 0.0.0.0:80 0.0.0.0:* users:(("nginx",pid=1234,fd=6))
```

```
tcp LISTEN 0 128 0.0.0.0:443 0.0.0.0:* users:(("chrome",pid=4567,fd=12))
```

## 3. Stop the Firewall Service

Temporarily halt the firewall process to isolate network rule restrictions.

Command:

```
sudo systemctl stop firewalld
```

Simulated Output:

(Command executes silently without errors)

#### 4. Confirm the Service Interruption

Verify that the service state transitioned from active to inactive.

Command:

```
sudo systemctl status firewalld
```

Simulated Output:

firewalld.service - firewalld - dynamic firewall daemon

Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; vendor preset: enabled)

Active: inactive (dead) since Thu 2026-08-20 18:27:00 IST; 4s ago

#### 5. Test Application Streaming

Launch or refresh Spotify Web in your browser to check if media audio streams resume playback normally.

#### 6. Restore Firewall Protection

Re-enable the security daemon immediately after testing to safeguard the operating system.

Command:

```
sudo systemctl start firewalld
```

Simulated Output:

(Command executes silently)

#### 7. Final State Verification

## MANAGE INSTALLED SERVICES

GUIDED BY : VIPUL MISTI

Ensure the firewall is operational again.

Command:

```
sudo systemctl is-active firewalld
```

Simulated Output:

```
active
```